Comparison of LSTM Classifier and Dynamic Classifier for HIF Detection

Shiqi Wu

March 12, 2026

1 Introduction

High-impedance faults (HIFs) in distribution systems occur when energized conductors come into contact with poorly conductive surfaces such as vegetation, asphalt, or sand. These faults are hazardous because they expose people and animals to electric shock and can trigger bushfires. However, their detection and protection remain challenging due to the small and irregular fault currents involved.

Most previous studies have focused on *detection*—that is, determining whether an HIF has occurred—rather than on *localization*. Yet, rapid and accurate localization is equally crucial for utilities to dispatch crews efficiently and restore service promptly. Unlike conventional low-impedance faults, HIF localization is considerably more difficult because the resulting fault current is often smaller than the background load current, leading to high uncertainty and weak fault signatures.

Recent advances in machine learning have demonstrated strong potential for improving fault diagnosis and localization. A growing body of research explores data-driven and hybrid approaches, including signal-processing-based feature extraction, synchronized measurements, and deep neural networks. Nevertheless, research dedicated specifically to HIF *localization* remains limited. Comprehensive reviews by Ghaderi *et al.* (2017) and Lopes *et al.* (2023) highlight that while machine learning has achieved notable progress in HIF *detection*, its application to localization is still in its early stage.

In this context, our goal is to investigate and compare two complementary strategies for HIF localization:

1. A purely data-driven model based on a Bidirectional Long Short-Term Memory (BiLSTM) network, which learns temporal dependencies directly from simulated time-series data.

2. A physics-informed dynamic classifier that formulates localization as an inverse problem over the power grid's state-space model.

The comparison between these two paradigms provides insight into the trade-off between predictive accuracy and physical interpretability. Specifically, we aim to assess whether the incorporation of physical structure—in the form of a dynamic model—can enhance interpretability without substantially compromising accuracy. The remainder of this report presents the dataset construction, model formulations, and comparative results, followed by a discussion of future directions for integrating data-driven and physics-based learning frameworks in power system fault analysis.

2 Dataset Construction

To evaluate and compare the two localization models, we generated a synthetic dataset based on simulations of the IEEE 34-node distribution feeder under high-impedance fault (HIF) conditions. The dataset construction process is summarized below.

2.1 Simulation Setup

Each simulation is performed in MATLAB using a standard HIF circuit topology consisting of anti-parallel diodes, variable resistors, and DC sources. Six candidate fault positions are selected along the feeder, as illustrated in Figure 1. For each fault position, we vary the arc resistance and breakdown voltage by sampling them from uniform distributions, capturing the stochastic nature of HIFs. The system is simulated at a sampling rate of 0.1 kHz, which provides sufficient temporal resolution for capturing fault transients.

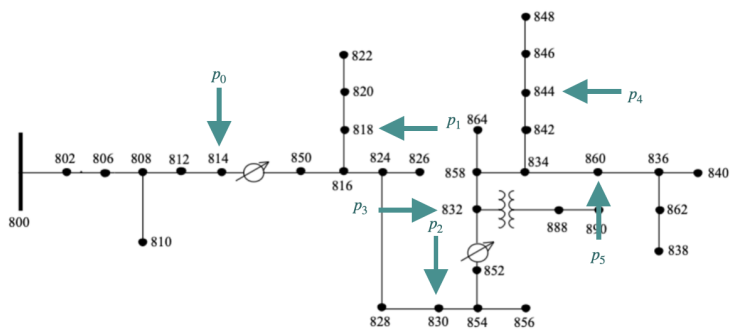

Figure 1: IEEE 34-node feeder used for HIF simulation. Six fault positions are labeled along the radial structure.

2.2 Trajectory Generation and Data Structure

For each fault location, we generate 20 independent trajectories, each of length 300 time steps, resulting in a total of 120 trajectories. The dataset is divided into 80% for training and 20% for validation. At each time step t, the system state vector x_t includes all bus voltages and branch currents, while u_t encodes the local HIF excitation.

Each trajectory is represented as a sequence of tuples:

$$(x, u, p),$$

where x denotes the system state sequence, u the corresponding fault-related input, and p the index of the fault position. Table 1 summarizes the meaning and dimensions of each component.

2.3 Preprocessing and Dimensionality Reduction

To improve computational efficiency and reduce noise, the high-dimensional state vectors are first standardized and projected to a lower-dimensional subspace using Principal Component Analysis

(PCA). We retain the first two principal components, which preserve more than 99% of the total variance. This preprocessing step ensures that both models—the BiLSTM and the dynamical sytstem-based classifier—operate on a consistent, compact representation of the system states.

The resulting dataset provides a controlled yet realistic testbed for comparing data-driven and physics-informed localization models under identical conditions.

Component	**Shape**	**Physical Meaning**
$x \in \mathbb{R}^{d_x}$	$(N_{\text{bus}} + N_{\text{branch}})$	Concatenated vector of all bus voltages $U_i(t)$ and branch currents $I_j(t)$ at time t.
$u \in \mathbb{R}^{d_u}$	$(2,)$	HIF input vector consisting of injected current and voltage at the fault location, i.e., $u = [I_{\text{HIF}}, U_{\text{HIF}}]$.
$p \in \{1, \ldots, 6\}$	scalar / one-hot $(6,)$	Fault position index among the six candidate buses, encoded as an integer or one-hot vector.

Table 1: Structure of dataset tuples (x, u, p), including shape and physical interpretation.

3 Problem Formulation

To simplify the initial exploration, we make several assumptions. First, the fault position p is considered constant throughout each trajectory. Second, we do not account for load variations or other external disturbances, focusing solely on the effect of the high-impedance fault (HIF). Moreover, we assume that an HIF is always present, so the task is not to detect its occurrence but rather to identify its location. Specifically, the objective is to classify the fault position p based on the observed system states x and inputs u. This naturally leads to a multi-class classification problem over time-series data, with six classes corresponding to the six possible fault positions.

In this report, we compare two approaches for HIF localization: a baseline Bidirectional Long Short-Term Memory (BiLSTM) classifier and a proposed dynamic classifier that leverages the underlying power grid dynamics. The BiLSTM serves as a black-box model that learns temporal patterns in the data, while the dynamic classifier incorporates physical insights into the system's behavior. We evaluate both methods on the same dataset and analyze their performance in terms of accuracy on different time windows.

4 Baseline: LSTM Classifier

We implement a Bidirectional Long Short-Term Memory (BiLSTM) network to classify the HIF position based on sequences of system states, serving as our baseline model. The BiLSTM architecture is chosen for its ability to capture temporal dependencies in both forward and backward directions. The input to the BiLSTM is a sequence of state vectors $X = (x_1, x_2, \ldots, x_T)$ over a specified window size T. The output is a probability distribution over the six possible fault positions. In section 4.1, we provide a detailed description of the BiLSTM architecture and its mathematical formulation. In section 4.2, we outline the specific architecture used, including the number of layers, hidden units, and training procedures.

4.1 Bidirectional LSTM (BiLSTM)

The Long Short-Term Memory (LSTM) network is a variant of recurrent neural networks (RNNs) designed to mitigate the vanishing and exploding gradient problems when modeling long-term temporal dependencies. A Bidirectional LSTM (BiLSTM) extends the standard LSTM by processing the input sequence in both forward and backward directions, thereby capturing past and future context simultaneously.

Formally, given an input sequence

$$X = (x_1, x_2, \ldots, x_T), \quad x_t \in \mathbb{R}^d,$$

a forward LSTM computes hidden states $(\overrightarrow{h}_1, \ldots, \overrightarrow{h}_T)$ sequentially from $t = 1$ to T, while a backward LSTM computes hidden states $(\overleftarrow{h}_1, \ldots, \overleftarrow{h}_T)$ from $t = T$ down to 1. The combined BiLSTM hidden state at time t is obtained by concatenation:

$$h_t = [\overrightarrow{h}_t; \overleftarrow{h}_t], \quad h_t \in \mathbb{R}^{2H},$$

where H is the dimensionality of the hidden state of each unidirectional LSTM.

The LSTM unit at each time step updates its cell state c_t and hidden state h_t through gated operations:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \tag{1}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \tag{2}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \tag{3}$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c), \tag{4}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{5}$$
$$h_t = o_t \odot \tanh(c_t), \tag{6}$$

where f_t, i_t, o_t are the forget, input, and output gates, respectively, $\odot$ denotes elementwise multiplication, and σ is the sigmoid activation function.

By incorporating both $\overrightarrow{h}_t$ and $\overleftarrow{h}_t$, BiLSTMs can leverage information from the entire sequence context, making them highly effective in sequence classification, fault detection, and other time-series tasks where both past and future dependencies matter.

4.2 Model Architecture and Training

The LSTM classifier architecture consists of the following components:

- **PCA Dimension Reduction:** The input state vector x_t is first reduced in dimensionality using Principal Component Analysis (PCA) to retain the most significant features while minimizing noise. This step helps improve training efficiency and model performance. In our experiments, we reduce the input dimension from 68 to 2.
- **BiLSTM Layer:** The reduced input sequence is fed into 2 Bidirectional LSTM layers with 128 hidden units in each direction. This layer captures temporal dependencies in the data.
- **Fully Connected Layer:** The final hidden states from the BiLSTM are passed through a feed forward fully connected layer with a ReLU activation function to introduce non-linearity.
- **Output Layer:** A linear layer produces the logits for the six fault positions.

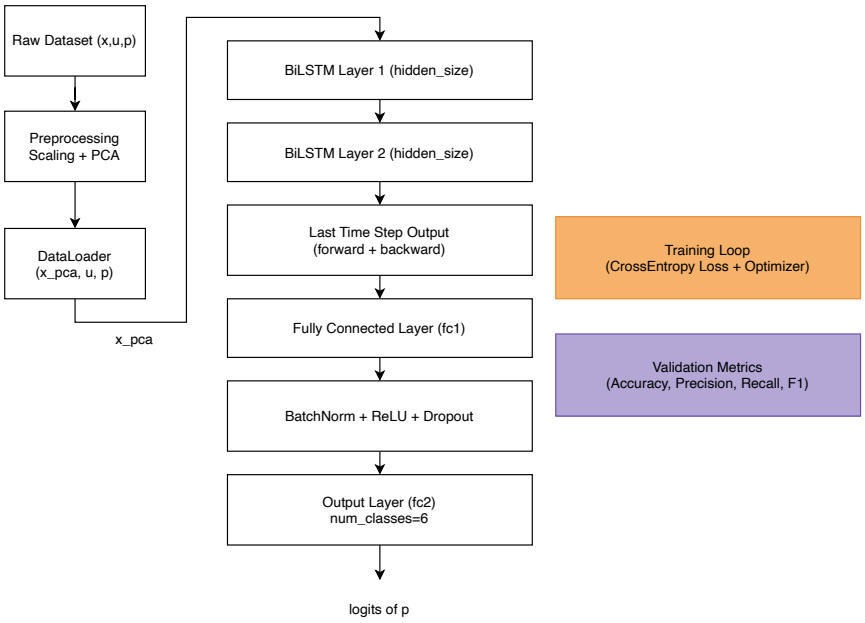

Figure 2: LSTM Classifier Architecture

The model architecture is depicted in Figure 2.

We use the CrossEntropyLoss function as the loss criterion:

$$\begin{aligned} \hat{y}_i = \text{softmax}(z_i) &= \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}, \\ \text{CrossEntropyLoss} &= -\sum_{i=1}^{C} y_i \log(\hat{y}_i), \end{aligned} \tag{7}$$

where C is the number of classes (6 fault positions), y_i is the true label (one-hot encoded), and $\hat{y}_i$ is the predicted probability for class i.

The model is trained using the Adam optimizer with a learning rate of 0.001 and weight decay of 0.00001. We use two NVIDIA 3090 GPUs for training, and the batch size is changed based on the window size to fit GPU memory. The training process is set to run for 3000 epoches, with early stopping based on validation accuracy to prevent overfitting.

5 Proposed: Dynamic Classifier

According to the analysis of the dynamical system of power grid network, we propose the dynamic classifier which formulates high-impedance fault (HIF) localization as an inverse problem on the power grid state-space model. Instead of directly learning a mapping from system trajectories to fault positions, we exploit the underlying dynamics to parameterize the effect of different fault locations. The diagram of the dynamic classifier is shown in Figure 3.

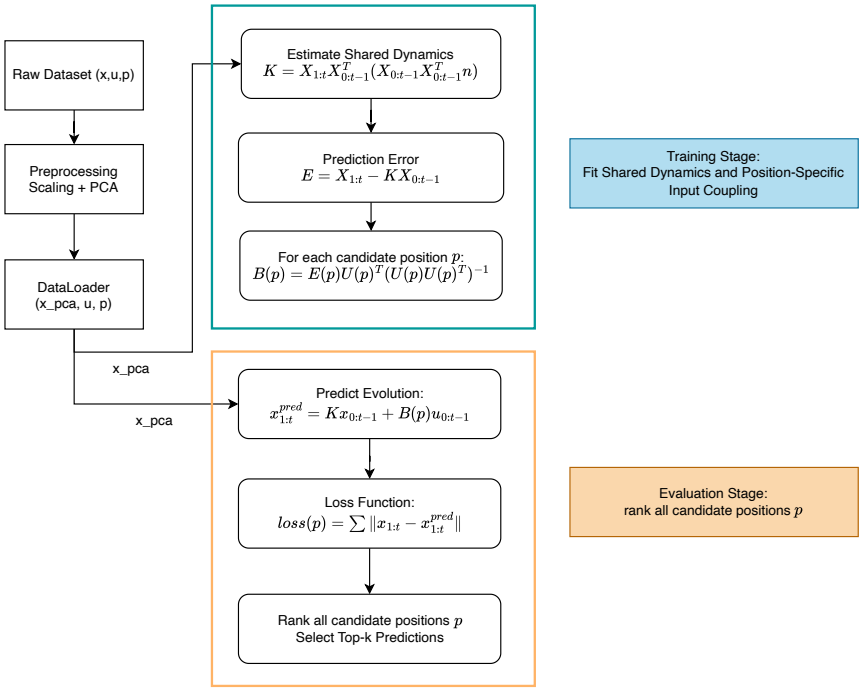

Figure 3: Dynamic Classifier Architecture

Parameter	Value / Description
Optimizer	Adam
Learning rate	0.001
Weight decay	1×10^{-5}
Learning rate scheduler	ReduceLROnPlateau (factor = 0.8, patience = 50 epochs)
Batch size	Adjusted according to window size and GPU memory
Window size	2000
Epochs	3000 (with early stopping based on validation accuracy)
Gradient accumulation	1
Momentum	0.9
Seed	42
Hardware	Two NVIDIA RTX 3090 GPUs
Model configuration	Input size: 2; Hidden size: 128; Num layers: 2; Bidirectional: True; Dropout: 0.2; Num classes: 6

Table 2: Training hyperparameters for the LSTM classifier.

5.1 Power Grid Dynamics

The power grid can be modeled as a set of interconnected nodes (buses) and edges (transmission lines), where each node represents a system component (e.g., generator, load) and each edge represents the power flow between components. The dynamics of the power grid are governed by a set of differential equations that describe the behavior of the system over time. These equations take into account various factors, including power generation, consumption, and transmission losses.

We model the power network dynamics in the standard state-space form

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{8}$$

where $x(t)$ denotes the system states (e.g., bus voltages and line currents), and $u(t)$ represents the external inputs to the system. The system matrix A is determined solely by the network topology and component parameters (L_i, R_i, C_i), and therefore remains invariant with respect to fault conditions.

In the presence of a high-impedance fault (HIF), the fault voltage and current $u = [I_{\text{HIF}}, U_{\text{HIF}}]$ can be regarded as an additional external input applied to the bus where the fault occurs. As a result, the HIF does not modify the intrinsic system dynamics captured by A, but only alters the way the input enters the system through the input matrix B. Concretely, different fault locations correspond to different columns of B, while the structure of A remains unchanged.

Therefore, in the state-space representation, the HIF location is encoded exclusively in the input matrix B, whereas the system matrix A is location-independent.

As a concrete example, consider the simple radial feeder shown in Fig. 4, consisting of buses $\{800, 802, 806\}$ connected by series (L, R) branches. Let the state vector be

$$x(t) = \begin{bmatrix} V_{802}(t) & V_{806}(t) & I_{800,802}(t) & I_{802,806}(t) \end{bmatrix}^\top, \tag{9}$$

where $V_{802}(t), V_{806}(t)$ are bus voltages and $I_{800,802}(t), I_{802,806}(t)$ are line currents. From Kirchhoff's

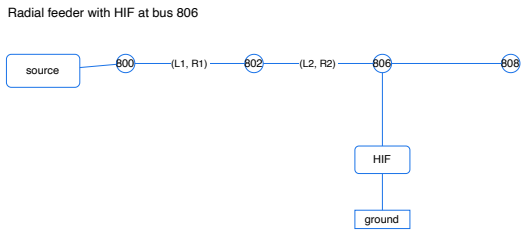

Figure 4: Simple radial feeder with HIF at bus 806

laws, the system dynamics can be written as

$$\dot{V}_{802}(t) = -\frac{1}{C_{802}}\Big(I_{800,802}(t) - I_{802,806}(t)\Big), \tag{10}$$

$$\dot{V}_{806}(t) = -\frac{1}{C_{806}}\Big(I_{802,806}(t) - \hat{i}_{\mathrm{HIF}}(t)\Big), \tag{11}$$

$$\dot{I}_{800,802}(t) = \frac{1}{L_1}\Big(V_{800}(t) - V_{802}(t) - R_1 I_{800,802}(t)\Big), \tag{12}$$

$$\dot{I}_{802,806}(t) = \frac{1}{L_2}\Big(V_{802}(t) - V_{806}(t) - R_2 I_{802,806}(t)\Big). \tag{13}$$

These equations can be cast in the compact state-space form

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{14}$$

where $u(t)$ collects the source excitation $V_{800}(t)$ and the HIF injection $\hat{i}_{\mathrm{HIF}}(t)$. The system matrix A depends only on network parameters $(L_1, R_1, L_2, R_2, C_{802}, C_{806})$ and is unaffected by the HIF. In contrast, the input matrix B takes the form

$$B = \begin{bmatrix} 0 & 0 \\ 0 & -1/C_{806} \\ 1/L_1 & 0 \\ 0 & 0 \end{bmatrix}, \tag{15}$$

where the first column corresponds to the source input $V_{800}(t)$ and the second column corresponds to the HIF injection.

It is evident that if the HIF occurs at a different bus, the structure of A remains unchanged, while only the column of B associated with the HIF input changes. Thus, the HIF location manifests solely through B in the state-space model.

5.2 Dataset Organization

We organize the dataset as a collection of triples:

$$(x, u, p),$$

where $x = x_{0:t-1}$ denotes the sequence of system states up to time $t-1$, $u = u_{0:t-1}$ is the corresponding sequence of fault-related inputs, and p is the position label of the HIF. The ground truth target is the next-step sequence $x_{1:t}$, which represents the observed system evolution. We first reduce the dimensionality of x using PCA, which is the same as the LSTM approach.

5.3 Shared Dynamics Estimation

We first solve for a shared system matrix K that captures the baseline dynamics of the power grid across all trajectories. Specifically, stacking the sequences across the dataset, we obtain

$$K = X_{1:t} X_{0:t-1}^T \left(X_{0:t-1} X_{0:t-1}^T \right)^{-1}, \tag{16}$$

where $X_{0:t-1}$ and $X_{1:t}$ are matrices concatenating all past and next-step states, respectively.

The residual error that is not explained by K is given by

$$E = X_{1:t} - K X_{0:t-1}. \tag{17}$$

5.4 Position-Specific Input Coupling

For each candidate HIF position p, we assume the effect of the fault is captured by an input coupling matrix $B(p)$. This matrix is estimated by regressing the unexplained error against the input sequence:

$$B(p) = E(p) U(p)^T \left(U(p) U(p)^T \right)^{-1}, \tag{18}$$

where $U(p)$ collects the input sequences associated with position p, and $E(p)$ is the corresponding error matrix. This formulation explicitly models how fault injections at different positions alter the grid dynamics.

5.5 Inverse Inference for Fault Localization

Given a candidate position p, the predicted system evolution is

$$x_{1:t}^{\text{pred}} = K x_{0:t-1} + B(p) u_{0:t-1}. \tag{19}$$

We compute a trajectory-level loss measuring the discrepancy between predicted and observed states:

$$\text{loss}(p) = \sum_{n=1}^{N} \left\| x_{1:t}^{\text{pred}} - x_{1:t} \right\|. \tag{20}$$

By ranking the losses over all candidate positions p, we infer the most likely fault location. The top-k positions are reported as final predictions.

5.6 Discussion

This framework separates shared grid dynamics, captured by K, from position-specific influences, represented by $B(p)$. Such a decomposition provides interpretability, as K and $B(p)$ respectively reveal the global behavior of the system and the localized effects of HIFs. It also offers flexibility: the model can adapt to different fault positions without requiring retraining of the entire system, since only $B(p)$ needs to be re-estimated. Moreover, by parameterizing $B(p)$ as Koopman operators with the fault position as a parameter, the framework has the potential to generalize to unseen fault locations.

Nonetheless, two limitations remain. First, the closed-form estimation of K and $B(p)$ is sensitive to measurement noise, which may reduce localization accuracy and make performance heavily dependent on the chosen window size; larger windows generally improve robustness. Second, the method assumes knowledge of the fault-related inputs u, which may not be directly measurable in practice. In real-world scenarios, these inputs would need to be estimated or inferred from the available system states x, inevitably introducing additional uncertainty into the classification process.

6 Parametric Koopman Operator

6.1 Parametric Koopman Operator

The parametric Koopman operator model aims to approximate nonlinear system dynamics through a learned nonlinear embedding $\Phi(x)$ and a set of class-dependent linear operators $\{K_c\}_{c=1}^{C}$. For each time step, the evolution in the latent space is assumed to follow a linear relation:

$$\Phi(x_{t+1}) \approx K_c\, \Phi(x_t), \tag{21}$$

where $K_c \in \mathbb{R}^{d\times d}$ represents the Koopman operator associated with class c.

The training objective minimizes the reconstruction residuals of the embedded trajectories:

$$\mathcal{L}_{\text{Koopman}} = \frac{1}{T-1}\sum_{t=1}^{T-1} \left\|\Phi(x_{t+1}) - K_c\Phi(x_t)\right\|_2^2. \tag{22}$$

Inference. Given a trajectory $x_{1:T}$, two inference strategies are applicable:

- **Residual-based classification.** For each candidate class c, compute the mean residual energy

$$e_c = \frac{1}{T-1}\sum_{t=1}^{T-1} \left\|\Phi(x_{t+1}) - K_c\Phi(x_t)\right\|_2^2, \tag{23}$$

  and predict the class with the smallest residual: $\hat{c} = \arg\min_c e_c$. Optionally, a softmax over $-e_c$ can be used to obtain class probabilities.

- **Simplex-based inference.** Alternatively, one can infer a probability vector $p \in \Delta^{C-1}$ by solving

$$\min_{p_c\geq 0,\, \sum_c p_c = 1} \left\|\Phi(x_{t+1}) - \sum_c p_c K_c \Phi(x_t)\right\|_2^2, \tag{24}$$

  which provides a soft probabilistic assignment across classes.

6.2 Residual Parametric Koopman Operator

The residual parametric Koopman operator extends the above formulation by decomposing each class operator into a shared global component and a class-specific residual term:

$$K_c = K_{\text{base}} + K_c^{\text{res}}. \tag{25}$$

The shared operator K_{base} captures the common dynamics among all classes, while the residual operators $\{K_c^{\text{res}}\}$ model class-dependent deviations.

Training procedure. Training is performed in two stages:

1. **Stage I (shared dynamics):** learn Φ and K_{base} by minimizing

$$\mathcal{L}_{\text{base}} = \frac{1}{T-1}\sum_{t} \left\|\Phi(x_{t+1}) - K_{\text{base}}\Phi(x_t)\right\|_2^2.$$

2. **Stage II (residual dynamics):** compute normalized residuals

$$r_t = \frac{\Phi(x_{t+1}) - K_{\text{base}}\Phi(x_t) - \mu}{\sigma},$$

and fit class-dependent operators K_c^{res} by minimizing

$$\mathcal{L}_{\text{res}} = \frac{1}{T-1}\sum_t \left\| r_t - K_c^{\text{res}}\Phi(x_t) \right\|_2^2.$$

Inference. Both inference strategies from the basic model can be applied:

- **Residual-based:**

$$e_c = \frac{1}{T-1}\sum_t \left\| (\Phi(x_{t+1}) - K_{\text{base}}\Phi(x_t)) - (\sigma K_c^{\text{res}}\Phi(x_t) + \mu) \right\|_2^2. \tag{26}$$

  The predicted class is $\hat{c} = \arg\min_c e_c$.

- **Simplex-based:** infer the probability vector $p \in \Delta^{C-1}$ such that

$$r_t \approx \sum_c p_c K_c^{\text{res}}\Phi(x_t), \quad p^* = \arg\min_{p_c \geq 0,\, \sum_c p_c = 1} \left\| r_t - \sum_c p_c K_c^{\text{res}}\Phi(x_t) \right\|_2^2. \tag{27}$$

This residual formulation captures both global and class-specific linear dynamics in a unified latent representation, leading to more stable and interpretable Koopman-based classification.

7 Results and Comparison

This section compares the performance of the purely data-driven BiLSTM classifier with the physics-based dynamic and Koopman-operator classifiers using the same training and validation datasets. We focus on quantitative accuracy, dependence on temporal window length, and interpretability of the learned dynamical representations.

7.1 Quantitative Performance Comparison

Figure 5 summarizes the classification accuracy across different window lengths. The BiLSTM classifier achieves a top-1 validation accuracy of approximately 70%, significantly outperforming all dynamic-system-based methods. Even under idealized conditions—where the dynamic model is provided with the true HIF current and voltage inputs—its accuracy only matches, but never exceeds, the BiLSTM baseline.

When the fault current and voltage are *unknown* and must be inferred indirectly, both Koopman-based methods (the standard parametric and the residual formulations) achieve only around 30-40% accuracy. This gap illustrates the strong dependence of dynamic approaches on the observability of physical inputs, as well as their sensitivity to noise and modeling uncertainty.

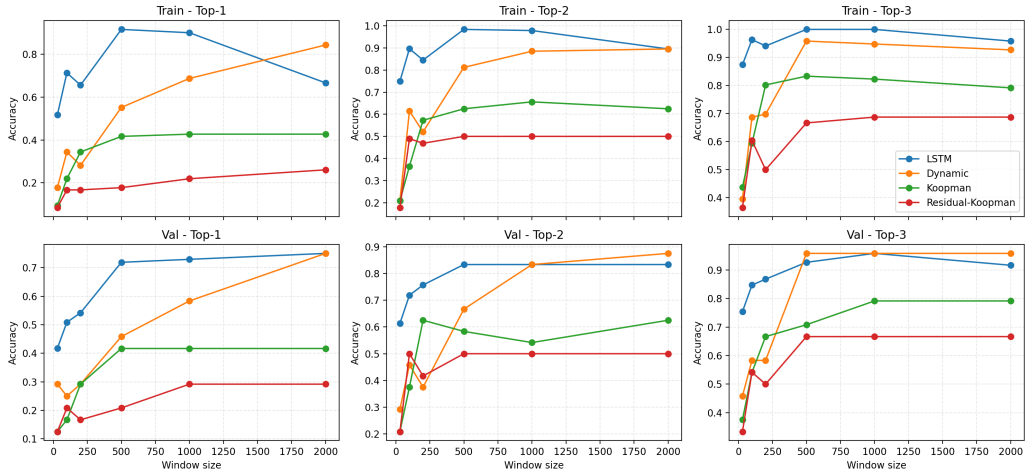

Figure 5: Comparison of LSTM and dynamic classifiers on training and validation datasets. The BiLSTM achieves the highest overall accuracy, while Koopman-based methods perform significantly worse when HIF inputs are unobserved.

7.2 Dependence on Window Length

A notable characteristic of the dynamic classifier is its strong dependence on the temporal window size T. As shown in Figure 5, its performance improves steadily with larger T, because longer trajectories provide more information for estimating the shared system operator K and the position-specific coupling $B(p)$.

However, this improvement only occurs when the positional effect on the system dynamics is *much larger* than the stochastic variation introduced by random simulation noise. When this condition is not met, the estimation of $B(p)$ becomes unstable, leading to poor localization accuracy. In contrast, the BiLSTM remains comparatively robust across window lengths, confirming that data-driven temporal embeddings can implicitly capture dynamic dependencies without explicit modeling of system matrices.

7.3 Koopman Operator Analysis

To interpret the learned dynamics, Figures 6–7 display the Frobenius distances between Koopman operators for different fault classes. The operators show distinct but subtle separations across fault positions, indicating that the method captures weak spatial signatures in the system response.

A closer look at the prediction error distributions (Figures 8 and 9) further reveals that the Koopman operators for "positive" and "negative" classes produce slightly different residual distributions. Although these differences are statistically consistent, they are often too small to yield reliable classification boundaries. This subtlety reflects the main limitation of the current Koopman-based formulation: while it captures qualitative differences in dynamics, the magnitude of these differences is insufficient for robust quantitative localization.

7.4 Discussion

In summary, the results demonstrate three key findings:

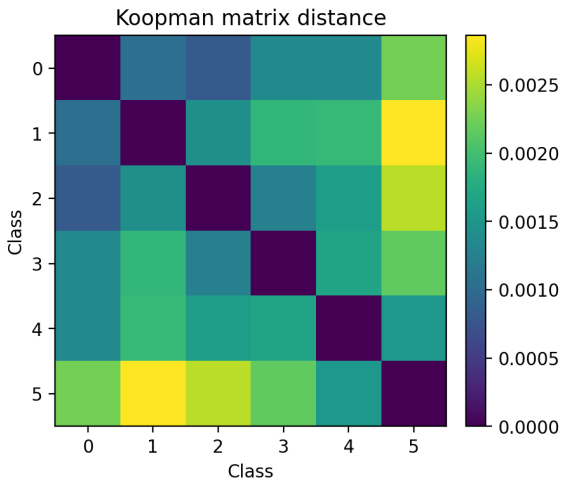

Figure 6: Frobenius norm of pairwise differences between Koopman operators K_c.

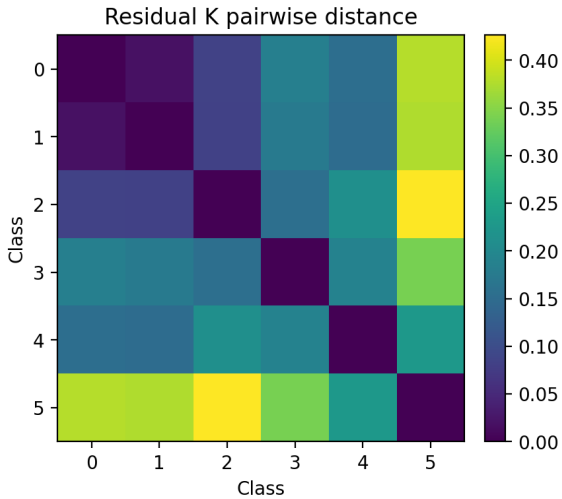

Figure 7: Frobenius norm of pairwise differences between residual Koopman operators K_c^{res}.

- **Performance gap:** Pure machine learning approaches, such as the BiLSTM, currently outperform dynamic-system-based methods by a large margin. Even the idealized dynamic model (with known HIF inputs) performs only on par with the BiLSTM, while Koopman variants without input knowledge drop to roughly 40% accuracy.
- **Window-size sensitivity:** The success of dynamic classifiers strongly depends on the evaluation window length, requiring that position-induced systematic variations dominate random fluctuations.
- **Weak Koopman separability:** Although the Koopman operators for different classes exhibit

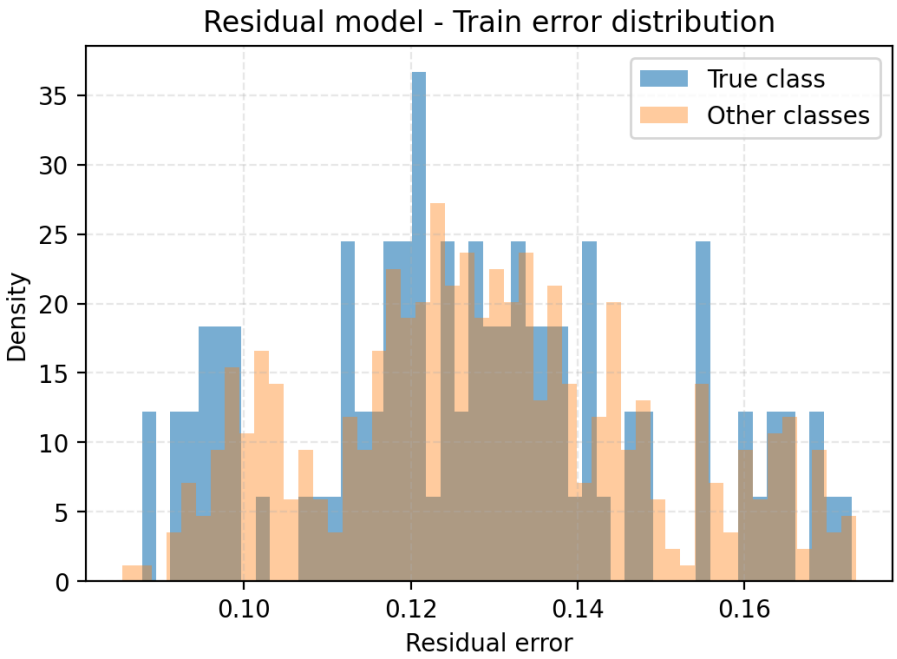

Figure 8: Distribution of prediction errors between Koopman-predicted and true trajectories for positive and negative classes on the training dataset.

minor distributional differences, these are not sufficiently distinct to ensure reliable classification, highlighting a key limitation of the current formulation.

These observations suggest that while physics-informed methods provide valuable interpretability, their practical utility for HIF localization remains limited without further enhancement in noise robustness and discriminative operator learning.

8 Conclusion and Future Plan

8.1 Conclusion

Our results show that purely data-driven approaches currently outperform physics-based methods for HIF localization. The BiLSTM classifier reaches about 70% validation accuracy, while the idealized dynamic model—assuming known HIF current and voltage—achieves only comparable performance. Without these inputs, the Koopman-based methods drop to roughly 40% accuracy. This confirms that dynamic formulations are highly sensitive to noise, window length, and input observability, even though they provide valuable physical interpretability.

8.2 Future Plan

Future work will focus on improving both accuracy and robustness of the dynamic framework:

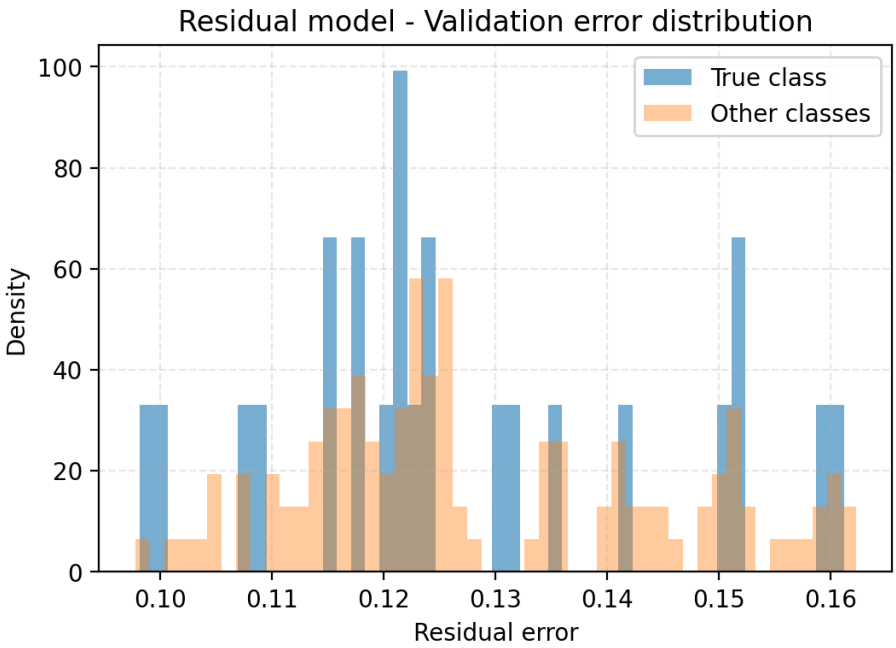

Figure 9: Distribution of prediction errors between Koopman-predicted and true trajectories on the validation dataset. Minor shifts between classes suggest weak separability of Koopman residuals.

1. **Joint identification and localization:** Extend the dynamic model to simultaneously detect whether an HIF exists and identify its position, enabling a fair comparison with the LSTM baseline on both tasks.

2. **Hybrid Koopman–LSTM approach:** Use an LSTM module as a temporal embedding or correction dictionary to enhance the separability of Koopman operators by learning longer temporal dependencies during training.

3. **Graph-based modeling:** Implement the position-dependent operator $B(p)$ with a Graph Neural Network (GNN) to represent the physical topology of the distribution network and improve location generalization.